Diego Alejandro Escobar Barahona

Junior Data Engineer | Machine Learning Researcher | Cloud Specialist

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Professional Summary

Junior Data Engineer & ML Researcher with enterprise internship experience at **GBM (IBM & Google Cloud Premier Partner)** and co-author in empirical credit risk research (OntoTab, 30,000 records). **Google Cloud Certified Associate Data Practitioner** with solid foundation in cloud ELT architectures on **Google Cloud (BigQuery, Dataform SQLX, Datastream CDC)**, integration tasks in **IBM InfoSphere DataStage, IBM DB2 LUW on Linux RHEL**, tabular ML modeling in **Python (XGBoost, Pandas, PyArrow)**, and **SQL**. Fluent in English (**EF SET C1 Advanced, 66/100, Listening C2 Proficient**).

Scientific Research & Publications

OntoTab: A Controlled Study of Ontology-Enriched Text for Tabular Credit Risk – *Co-author (2nd of 7), GBM* 2026

- Executed baseline experiments (Activity E01) evaluating **30,000 credit records** (UCI Taiwan, 22.12% default rate) across Logistic Regression, XGBoost, and MLP using **5×3 stratified nested cross-validation** and **2,000 paired bootstrap iterations** via CLI workflows.
- Analyzed performance metrics **supporting the finding that structured numerical representations outperformed frozen text embeddings in the evaluated experimental setting** (**0.5596 AP with XGBoost** vs 0.5069 AP with R5 text representation).

Experience

Data Engineer Intern (Digital Solutions Specialist), GBM (IBM & Google Cloud Premier Partner) – San Salvador July 2026 – Oct 2026

- Designed an **ELT cloud replication architecture proposal** on **Google Cloud** (**BigQuery, Dataform SQLX, Datastream CDC, Cloud Storage**) delivering **Lucidchart** data flow diagrams and **SQL DDL** scripts.
- Developed and executed **hands-on ETL integration tasks under technical mentorship** in **IBM InfoSphere DataStage** (Transformers, Lookups, Joins, Merge, Funnel) across enterprise test datasets, querying **IBM DB2 LUW** on **Linux RHEL**.
- Assisted in formulating GCP cloud governance frameworks and FinOps budget validations (least-privilege **IAM, Service Accounts**, credential rotation) and infrastructure sizing in **Google Cloud Pricing Calculator**.
- Built a Python CLI automation tool under **Spec-Driven Development (SDD)** to compile corporate technical documentation to PDF, assisting team members with **Git/SSH** workflows.

Projects

FERROSTATS — Cloud-Native Medallion Data Lakehouse & Streaming Pipeline on GCP GitHub | Looker Studio BI | Live App

- Architected a cloud-native **Medallion Lakehouse (Bronze / Silver / Gold)** in **Google BigQuery** with **Dataform SQLX**, structuring 7 raw landing tables into curated Silver views and **3 Gold Star Schema analytical marts** (`fct_daily_sales`, `mart_inventory_pareto_abc`, `dim_products`).
- Formulated functional specifications under **Spec-Driven Development (SDD)** for 80/20 Pareto ABC inventory segmentation, Daily Burn Rate, and DOI, powering a 3-page executive dashboard in **Looker Studio** with analytical SQL window functions.
- Engineered a decoupled OLTP-to-OLAP streaming ingestion pipeline achieving **<50 ms** latency into BigQuery Bronze via REST API with ACID concurrency constraints, achieving a permanent **$0.00/mo FinOps architecture on GCP** via **Cloud Run**.

Skills

Cloud & Warehouses: Google Cloud Platform (BigQuery, Dataform SQLX, Datastream CDC, Cloud Storage, Cloud Run, Artifact Registry, Looker Studio, IAM Service Accounts, Cloud Logging/Monitoring, FinOps)

Machine Learning & Modeling: 5x3 Nested Cross-Validation, Bootstrap Resampling, XGBoost, Logistic Regression, BGE-M3 Embeddings, Average Precision (AP), Medallion Architecture, Star Schema

Languages & Querying: SQL (GoogleSQL, PostgreSQL 18, Window Functions, CTEs, Anti-joins, DDL/DML), Python 3.11 (Pandas, PyArrow, NumPy), Bash, PowerShell, Docker, Linux RHEL, Git/SSH

Languages: Spanish (Native), English (C1 Advanced — EF SET 66/100, Listening C2 Proficient)

Certifications

Google Cloud Certified — Associate Data Practitioner (Cert ID: 23e3584a246f4c149f26fea03cebcbbb) Oct 2026 – Oct 2029

Google Cloud Skill Badges: Derive Insights from Data | Prepare Data for Looker Dashboards and Reports Aug 2026

EF SET English Certificate — C1 Advanced (66/100 global • Listening 75 C2 • Speaking 69 C1) Jan 2026

Code in Place (Python Programming) — Stanford University 2026

Education

University of the People — Pasadena, California, USA Admitted | Term 3, 2027

Bachelor of Science in Computer Science WSCUC-accredited institution | 100% Online

Technical Baccalaureate in Software Development – Instituto Nacional San Luis | *GPA: 9.5/10.0* Expected December 2026

- **Key Curriculum:** Algorithmic Logic, Relational Databases (SQL), Data Structures, Web Architecture & Software Engineering.